FTE4312: Blockchain and Decentralized Applications

Homework Assignment

Instructions: Answer in English. Show calculations for Part C. Keep explanations concise.
Note: Use the simplified rules stated in each question (toy models).

Part A — True / False (10 questions)

Write **T** or **F** for each statement, and **briefly explain your reasoning**.

1. In a constant-product AMM (like Uniswap v2), if swap fees are kept inside the pool, the value of k typically increases after a swap. _____

   Explanation: _____

2. A liquidity provider can add liquidity to a Uniswap v2 pool in any token ratio they want and still receive LP tokens without any leftover. _____

   Explanation: _____

3. In a constant-product AMM, larger trades (as a fraction of reserves) generally cause larger slippage. _____

   Explanation: _____

4. In a lending protocol, the liquidation threshold is always the same as the maximum borrow limit. _____

   Explanation: _____

5. A common definition of utilization rate in a lending pool is: utilization = total borrowed / total supplied. _____

   Explanation: _____

6. A flash loan can be repaid in a later block as long as it is repaid before 1 hour passes. _____

   Explanation: _____

7. If an AMM price differs from the external market price, arbitrageurs are incentivized to trade until prices move closer. _____

   Explanation: _____

8. In a sandwich attack, the attacker's back-running transaction is placed before the victim's transaction. _____

   Explanation: _____

9. Liquidators in DeFi lending protocols are typically rewarded via a liquidation bonus (a discount on the collateral). _____

   Explanation: _____

10. Ethereum's move to Proof-of-Stake completely eliminates MEV (Miner/Maximal Extractable Value). _____

    Explanation: _____

Part B — Multiple Choice (10 questions)

Choose **ONE** answer (A–D) for each question, and **briefly explain your reasoning**.

1. A constant-product AMM has reserves 50 ETH and 100,000 USDC. Ignoring fees, the spot price of 1 ETH in USDC is:

   A. 2,000
   B. 0.0005
   C. 5,000
   D. Cannot be determined

   Explanation: _____

2. Which statement best describes impermanent loss (IL) for a Uniswap v2 liquidity provider (ignoring fees)?

   A. IL is always positive (LP always beats holding).
   B. IL occurs when the relative price changes; LP value can be lower than simply holding the tokens.
   C. IL only happens when trading fees are nonzero.
   D. IL is the gas fee paid by the LP.

   Explanation: _____

3. In an over-collateralized DeFi lending market, which parameter primarily limits how much you can initially borrow against collateral?

   A. Collateral factor (max LTV)
   B. Liquidation bonus
   C. Block gas limit
   D. Merkle root

   Explanation: _____

4. A lending pool has total supplied = 1,000,000 DAI and total borrowed = 600,000 DAI. Utilization is:

   A. 40%
   B. 60%
   C. 140%

D. 1,600%

Explanation: _____

5. A simple linear borrow-rate model is: BorrowAPR = base + slope × utilization. If base = 2% and slope = 20%, and utilization = 60%, then BorrowAPR is:

   A. 4%
   B. 12%
   C. 14%
   D. 22%

   Explanation: _____

6. In Uniswap v2, with fee rate f (e.g., $f = 0.003$), a common formula for amountOut is:

   A. $\text{amountOut} = \dfrac{\text{amountIn} \cdot \text{reserveOut}}{\text{reserveIn} + \text{amountIn}}$
   B. $\text{amountOut} = \dfrac{\text{amountIn}(1-f)\text{reserveOut}}{\text{reserveIn} + \text{amountIn}(1-f)}$
   C. $\text{amountOut} = \dfrac{\text{amountIn} \cdot \text{reserveOut}}{\text{reserveIn}}$ (no slippage)
   D. $\text{amountOut} = \text{reserveOut} - \text{reserveIn}$

   Explanation: _____

7. A borrower becomes eligible for liquidation when:

   A. Their health factor drops below 1 (or their collateral ratio drops below the threshold).
   B. They have ever borrowed before.
   C. They repay part of their debt.
   D. Utilization in the pool is 0%.

   Explanation: _____

8. If Pool A offers ETH cheaper than Pool B (in USDC per ETH), a profitable arbitrage sequence is usually:

   A. Buy ETH on Pool B, sell ETH on Pool A
   B. Buy ETH on Pool A, sell ETH on Pool B
   C. Buy USDC on both pools
   D. No arbitrage is possible in AMMs

   Explanation: _____

9. If a flash-loan-based arbitrage does not repay the flash loan by the end of the transaction:

   A. The protocol will send an email reminder.
   B. The loan is rolled into a long-term debt position automatically.
   C. The entire transaction reverts (fails) atomically.
   D. Only the last step reverts; earlier steps remain.

   Explanation: _____

10. Suppose a stablecoin (e.g., DAI) trades at 0.98 USDC on a DEX, while it can be sold elsewhere for approximately 1.00 USDC. A typical arbitrage action is to:

    A. Sell DAI on the DEX (because it is overpriced).

    B. Buy DAI on the DEX and sell it elsewhere for approximately 1.00 USDC.

    C. Do nothing; arbitrage is impossible with stablecoins.

    D. Burn DAI to increase its price.

    Explanation: _____

Part C — Computational Problems (7 problems)

C1) Swap with fee and slippage

A Uniswap v2-style pool has reserves: 50,000 USDC and 25 ETH. The swap fee is 0.30% (i.e., $f = 0.003$). A trader swaps 5,000 USDC for ETH.

a) Compute the amount of ETH the trader receives.

b) Compute the new reserves of the pool after the swap.

c) Compute the spot price (USDC per ETH) before and after the swap.

d) If the trader sets 1% slippage tolerance based on the expected output, what is the minimum ETH output they will accept?

C2) Liquidity providing and impermanent loss

A pool starts with 100 ETH and 200,000 DAI (so the initial price is 2,000 DAI/ETH). Alice deposits 10 ETH and 20,000 DAI, and receives 10% of the LP shares. Assume no fees for this problem.

a) After an external price change, arbitrage brings the pool price to 2,500 DAI/ETH. Compute the new pool reserves (ETH and DAI).

b) How many ETH and DAI does Alice own via her LP shares after rebalancing?

c) Compute Alice's LP position value (in DAI) and compare it to the value of simply holding 10 ETH + 20,000 DAI. What is the impermanent loss percentage?

C3) Cross-pool arbitrage

Two ETH/USDC pools (no fees) have the following reserves:

- Pool A: 100 ETH and 200,000 USDC
- Pool B: 100 ETH and 220,000 USDC

An arbitrageur starts with 5,000 USDC and performs: (i) swap USDC→ETH on Pool A, then (ii) swap that ETH→USDC on Pool B.

a) How much ETH does the arbitrageur get from Pool A?

b) How much USDC do they receive from Pool B? What is the profit in USDC?

c) Compute the spot price (USDC/ETH) in each pool after the arbitrage sequence.

C4) Lending health factor and liquidation math

Alice deposits 10 ETH as collateral. Initially, ETH = 2,000 DAI/ETH. The collateral factor is 75% and the liquidation threshold is 80%. Alice borrows 14,000 DAI. Later, ETH drops to 1,650 DAI/ETH.

Define health factor HF = (collateral value × liquidation threshold)/debt value.

a) Compute Alice's health factor after the price drop. Is she liquidatable?

b) What is the maximum debt (in DAI) Alice can have at ETH = 1,650 without being liquidated (HF $\geq$ 1)?

c) How much debt must be repaid (in DAI) to restore HF to 1, assuming collateral amount stays the same?

d) Assume a liquidator repays 5,000 DAI of Alice's debt. If the liquidation bonus is 5%, how much ETH can the liquidator seize? (Assume no close factor limits.)

C5) Interest rates from utilization

A DAI lending pool has total supplied = 1,000,000 DAI and total borrowed = 600,000 DAI. Utilization u = borrowed/supplied.

BorrowAPR = base + slope $\times$ u, where base = 2% and slope = 20%. The reserve factor is 10%.

SupplyAPR = BorrowAPR $\times$ u $\times$ (1 − reserve factor). Assume simple interest (no compounding).

a) Compute utilization u, BorrowAPR, and SupplyAPR.

b) A lender supplies 50,000 DAI for 30 days. How much interest do they earn (in DAI)?

c) A borrower borrows 10,000 DAI for 30 days. How much interest do they owe (in DAI)?

C6) Sandwich attack

Consider an AMM DEX with constant product $x \cdot y = k$ and no swap fee. The pool starts with 100 ETH and 300,000 USDT (so k = 30,000,000).

A victim wants to swap 20 ETH for USDT. An attacker front-runs with 5 ETH, swapping ETH for USDT. The attacker then back-runs by swapping all of that USDT back to ETH.

a) How many USDT does the attacker get from the front-run?

b) How many USDT does the victim get (with the attacker present)?

c) How many USDT would the victim get if there were no attacker (victim swaps 20 ETH

directly on the initial pool)?

d) Assuming the attacker back-runs by swapping all their USDT back to ETH, how many ETH do they end with? What is the attacker's profit in ETH?

C7) Liquidation trigger price

Bob deposits 2 ETH as collateral and borrows 2,500 DAI. The liquidation threshold is 80%. Ignore interest accrual.

a) Derive the ETH price (in DAI/ETH) at which Bob becomes liquidatable (HF = 1).

b) If the collateral factor is 75% and ETH initially is 2,000 DAI/ETH, what is the maximum amount of DAI Bob could have borrowed initially?

c) If ETH later falls to 1,400 DAI/ETH (and debt is still 2,500 DAI), compute Bob's health factor and the minimum debt repayment needed to bring HF back to 1 (keeping collateral fixed).